

User Study

Instructions for Grading. **PLEASE READ CAREFULLY.**

Overview

Imagine you are a user interacting with a chatbot service (e.g., Gemini, ChatGPT, or DeepSeek) for help with programming tasks. On each page, you will see:

- **A Question:** This could be a vague user query or a code snippet requiring clarification or completion. Questions containing code will be displayed in code blocks for clarity.
- **Two Responses:** labeled Answer A and Answer B

Your Task

Please evaluate Answer A and Answer B based on the following criteria:

- **Precision and Focus:** How directly the response targets the question's core issue, using clear guidance, examples, or code.
- **Contextual Fit:** How well the response matches the user's programming goal (e.g., debugging, feature implementation).
- **Answer Faithfulness:** How effectively the response provides actionable steps or code to achieve the user's desired outcome.
- **Correctness:** Correctness in LLM responses refers to the accuracy and factual validity of the information, code, or guidance provided, ensuring it is free from errors and aligns with the user's query and intended outcome.

How to Rate

For each question, you will provide two types of ratings:

1. Preference Rating (1–5 scale):

- **"1"**: Answer A is better, based on the criteria above.
- **"5"**: Answer B is better, based on the criteria above.
- **"3"**: Both answers are equally good, based on the criteria above.

Note:

- **Incomplete Responses:** Some responses may appear cut off due to a 256-word limit imposed on the chatbots. Please grade only the content provided, ignoring any abrupt endings or non-English text (e.g., stray characters).
- **Code Formatting:** Questions and responses containing code will be presented in code blocks or artifact tags for clarity. Ensure you review these carefully for correctness and relevance.
- **Consistency:** Try to apply the same standards across all evaluations to ensure fair comparisons.

Question 1

I need assistance with a function.

Answer A:

Okay, I'd love to! Please tell me about the function you need assistance with. To help me understand and give you the best support, please provide as much detail as possible, including:

1. **What is the function supposed to do?** (Describe the purpose of the function in plain English)
2. **What is the input to the function?** (What data does it receive? Be specific - data types, examples)
3. **What is the expected output of the function?** (What data does it return? Be specific - data types, examples)
4. **What have you tried so far?** (Have you written any code? If so, please share it. Even if it's not working, it helps me understand where you're stuck.)
5. **What specific problem are you encountering?** (Are you getting an error message? Are you unsure how to implement a particular part of the code? Are you struggling with a specific concept?)
6. **Are there any constraints or limitations?** (e.g., "I need to use a specific library," or "I need to keep the code concise.")

The more information you give me, the better I can help.

Answer B:

“Extra context:

User Question: *I need assistance with a function.*

AI clarification question: *Please share the function code you're working on. That will help me understand what you're trying to accomplish and how you're approaching the problem.*

User follow up response: *I'm working on a Python function to validate email addresses, but it's not catching invalid formats.”*

AI Final answer:

Great! Let's craft a function to validate email addresses in Python. Here's a well-commented function that uses a regular expression to check for a basic email pattern:

```
Python
import re

def validate_email(email):
    """Validates an email address using a regular expression."""
    pattern = r"^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$"
    return bool(re.match(pattern, email))
```

```
# Example usage:  
print(validate_email("test@example.com"))    # True  
print(validate_email("invalid-email"))        # False
```

This function returns **True** for valid emails and **False** for invalid ones. If you need stricter validation or want to handle edge cases, let me know!

How to rate:

"1": strongly prefer answer A

"5": strongly prefer answer B

"3": both answers are equally good

Precision and Focus (1-5):5

Contextual Fit (1-5):5

Answer Faithfulness (1-5):5

Correctness (1-5):5

Question 2

Python

```
import requests
# TODO: Ensure headers are set for API request
response = requests.get('https://api.example.com/data')
print(response)
```

Answer A:

“Extra context:

User Question:

Python

```
import requests

# TODO: Ensure headers are set for API request
response = requests.get('https://api.example.com/data')
print(response)
```

AI clarification question: Please clarify what you want to achieve with this code snippet. What API endpoint are you trying to access and what outcome are you hoping to get from the API response?

User follow up response: I'm trying to fetch user data from a custom API, expecting a JSON response with user details.”

AI Final answer:

You're absolutely right to clarify! If you're expecting a JSON response with user details, you should set the appropriate headers and parse the response. Here's an example:

JavaScript

```
import requests

headers = {'Accept': 'application/json'}
response = requests.get('https://api.example.com/data', headers=headers)

if response.ok:
    user_data = response.json()
    print(user_data)
else:
    print(f"Request failed: {response.status_code}")
```

This ensures the API knows you expect JSON, and it prints the parsed user data.

Answer B:

```
JavaScript
import requests

# TODO: Ensure headers are set for API request
response = requests.get('https://api.example.com/data')
print(response)
```

Explanation:

1. **import requests**: Imports the `requests` library.
2. **response = requests.get(...)**: Sends an HTTP GET request to the specified URL.
3. **print(response)**: Prints the response object to the console.

How to rate:

"1": strongly prefer answer A

"5": strongly prefer answer B

"3": both answers are equally good

Precision and Focus (1-5):1

Contextual Fit (1-5):1

Answer Faithfulness (1-5):1

Correctness (1-5):1

Question 3

I'm facing issues with deploying my website.

Answer A:

Okay, let's tackle this! Deploying a website can be tricky, and it's frustrating when things don't go smoothly. To help me understand the situation and offer the best advice, I need a little more information. Let's break this down into categories.

1. What kind of website are you deploying? (e.g., WordPress, Shopify, React, Node.js, Laravel, custom-built)
2. What's the size of the website? (e.g., small blog, e-commerce store, portfolio, complex web app)
3. What's the primary platform you're using? (e.g., AWS, Google Cloud, DigitalOcean, self-hosted server)
4. What's the specific problem you're experiencing? Be as detailed as possible.
 - Error Messages: Can you share any error messages you're seeing?
 - Deployment Process: How are you trying to deploy?
 - What have you tried so far?

Answer B:

“Extra context:

User Question: I'm facing issues with deploying my website.

AI clarification question: Could you tell me which part of the deployment process is causing problems? For example, are you experiencing errors related to server access, permissions, or something else?

User follow up response: I'm deploying to Heroku, and I'm getting a 'Permission denied' error during deployment.”

AI Final answer:

A 'Permission denied' error during deployment to Heroku is common. Here's how to troubleshoot:

1. **Check Git remote:** Make sure your Git remote URL is correct and you have access.
2. **Heroku CLI login:** Run `heroku login` in your terminal to ensure you're authenticated.
3. **SSH keys:** If using SSH, ensure your public key is added to your Heroku account (`heroku keys:add`).
4. **File permissions:** Make sure files in your repo are not set to restricted permissions.

If you provide the full error message or your deployment command, I can give more targeted help!

How to rate:

"1": strongly prefer answer A

"5": strongly prefer answer B

"3": both answers are equally good

Precision and Focus (1-5):5

Contextual Fit (1-5):5

Answer Faithfulness (1-5):5

Correctness (1-5):5

Question 4

I'm confused about this library.

Answer A:

“Extra context:

User Question: *I'm confused about this library.*

AI clarification question: *Could you tell me what library you're referring to and what specifically you're finding confusing about it? For example, are you unsure about its purpose, how it works, or what features you're struggling with?*

User follow up response: *I'm confused about how to use the Matplotlib library in Python to create plots.”*

AI Final answer:

Matplotlib is a popular Python library for creating visualizations. To make a simple plot, you can use:

```
Python
import matplotlib.pyplot as plt

x = [1, 2, 3, 4]
y = [10, 20, 25, 30]
plt.plot(x, y)
plt.xlabel('X-axis label')
plt.ylabel('Y-axis label')
plt.title('My First Plot')
plt.show()
```

If you want to create a specific type of plot (bar, scatter, etc.) or customize it, let me know what you're trying to achieve!

Answer B:

Please tell me about the library! I need some information to understand what you're confused about. 😊

- What library are you referring to?
- What specifically are you confused about?
- Can you give me a little more context?

The more information you give me, the better I can understand your confusion and provide a helpful explanation.

How to rate:

"1": strongly prefer answer A

"5": strongly prefer answer B

"3": both answers are equally good

Precision and Focus (1-5):1

Contextual Fit (1-5):1

Answer Faithfulness (1-5):1

Correctness (1-5):1

Question 5

I need assistance with debugging.

Answer A:

“Extra context:

User Question: *I need assistance with debugging.*

AI clarification question: *What programming language are you using? (e.g., Python, JavaScript, C++, etc.)*

User follow up response: *I'm debugging a Python script, and I'm getting an 'IndexError: list index out of range'.*”

AI Final answer:

An `IndexError: list index out of range` in Python means your code is trying to access a list element at an index that doesn't exist. To fix this:

- Check the length of your list before accessing elements.
- Print the list and the index you're using to debug.
- Use `len(list)` to ensure your index is within bounds.

Example:

```
Python
my_list = [1, 2, 3]
for i in range(len(my_list)):
    print(my_list[i]) # This is safe
```

If you share your code snippet, I can help pinpoint the exact problem!

Answer B:

Okay, I'm here to help! Let's tackle this. To give you the best assistance, I need a bit more information. Let's start with some basic questions:

- What is the code you're working on? Please share the relevant code snippet.
- What is the problem you're trying to solve?
- What have you already tried?
- What language are you using?

The more information you provide, the better I can help.

How to rate:

"1": strongly prefer answer A

"5": strongly prefer answer B

"3": both answers are equally good

Precision and Focus (1-5):1

Contextual Fit (1-5):1

Answer Faithfulness (1-5):1

Correctness (1-5):1